

MMD Generative Adversarial Networks

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Standard GANs vs. MMD GANs

The Shared Foundation

Both frameworks operate on a simultaneous optimization process solving a standard **min-max objective** between a generator G and a critic/discriminator D :

$$\min_G \max_D \mathcal{L}(G, D)$$

Standard GANs (JS / Wasserstein)

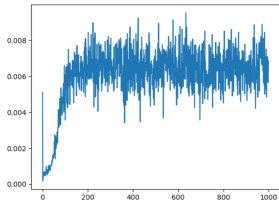
- Critic D is parameterized as a deep neural network updated iteratively via backpropagation.
- Requires highly sensitive alternating optimization steps.
- **No analytical expression** exists for the optimal discriminator at each discrete step.

MMD GANs (Maximum Mean Discrepancy)

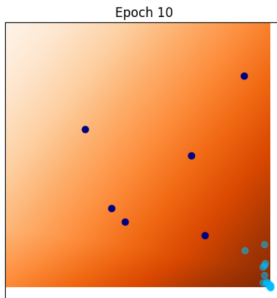
- The critic leverages the Reproducing Kernel Hilbert Space (RKHS) framework.
- **Explicit Closed Form:** Yields a specific analytical formula for the optimal discriminator given a fixed generator G .
- **Numerical Computation:** The theoretical closed form is directly estimated using empirical means over mini-batches.

Standard GAN vs. WGAN: Training & Stability Issues

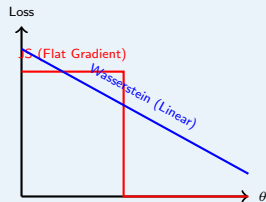
Discriminator loss in Weight clipping setting



Handling balance of updates G/D



Visualizing Divergences



Top line: JS divergence creates flat gradient plateaus.

Bottom line: WGAN maintains continuous, meaningful updates.

MMD GAN Training on 2D Synthetic Data

Empirical MMD Formulations Used

$$\widehat{\text{MMD}}^2(\mathcal{X}, \mathcal{Y}) = \frac{1}{n^2} \sum_{i,j} k(x_i, x_j) - \frac{2}{nm} \sum_{i,j} k(x_i, y_j) + \frac{1}{m^2} \sum_{i,j} k(y_i, y_j)$$

Epoch 95 - Witness Function

